



GROUP 05 · THREE ALPHA

MedReview

A proof-verified review platform for hospitals, doctors & medical services

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INTRODUCTION

Trusted - Verified

MedReview is a secure, web-based platform that lets patients rate and review hospitals, doctors, and medical services—with proof-backed verification to stop fake or manipulated reviews.



Trust deficit in healthcare reviews

Patients struggle to choose providers because existing platforms allow anonymous, unverified feedback.



Proof-verified, multi-dimensional ratings

Every review requires evidence - a bill, appointment number, or document before it's approved.



A single web-based directory

One relational system covering hospitals, doctors, and medical services in one place.



Every review needs proof

Bill upload

Appointment number

Supporting document

What's broken in healthcare reviews today



No proof of visit

Anonymous, unverified feedback with no evidence an actual visit occurred.



Manipulated ratings

Fake positive reviews boost a clinic's image; fake negative ones damage a doctor's standing.



One flat star rating

A single 5-star scale missespunctuality, staff behavior, knowledge, bedside manner.



Shifting affiliations

Doctors practice across multiple hospitals; static directories can't keep up.



Weak moderation

No structured review process risks leaking sensitive patient information.



No audit trail

No centralized log of admin decisions or security events for compliance.

OBJECTIVES

What each side of the platform is built to do



Admin View

- Verify and moderate submitted reviews (approve/reject with notes)
- Manage the hospital, doctor & medical service directory
- Handle reports of spam, offensive content, or violations
- Monitor activity through a secure, auditable system log



Student / Patient View

- Search and browse verified providers by specialty or city
- Submit ratings backed by real proof
- Save and bookmark preferred providers for later
- Vote on review helpfulness and report bad content

General goal

Build patient trust through verification, multi-dimensional quality metrics, accurate doctor-hospital data, and HIPAA-style compliance.

FEATURES

Admin controls & security backbone

ADMIN FEATURES

- ✓ Create / update / delete hospitals, doctors, specialties, services
- ✓ Approve or reject pending reviews with internal notes
- ✓ Manage doctor-hospital affiliations
- ✓ View and act on user-submitted reports
- ✓ Access system logs for audit monitoring
- ✓ Manage user accounts



Security & Backend

Password hashing with bcrypt

Role-based access control via middleware

Input validation with Zod

Relational integrity via Prisma ORM + PostgreSQL

FEATURES

What a patient can do on the platform



Secure JWT auth

Register and log in securely with JWT- based authentication.



Browse the directory

Explore hospitals, doctors, and medical services.



Multi-category reviews

Rate staff, punctuality, knowledge, and bedside manner separately.



Upload proof

Attach a bill, appointment number, or file with every review.



Save listings

Bookmark providers for quick access later.



Vote helpfulness

Mark other reviews as helpful or not helpful.



Report content

Flag reviews that look fake, offensive, or non-compliant.

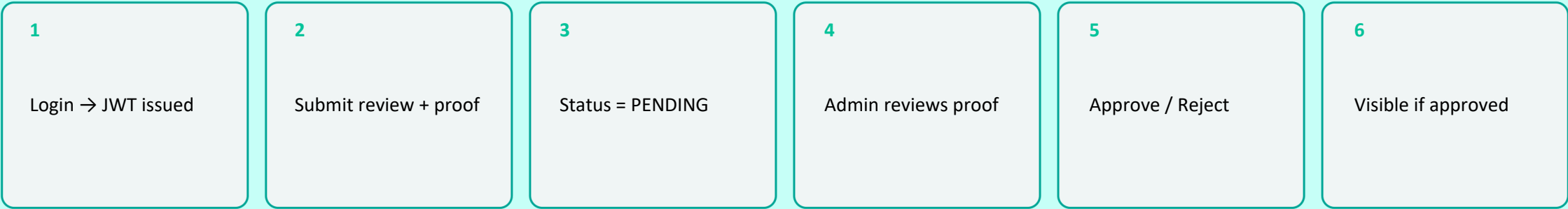
Edge cases the schema had to solve

- | | |
|----------------------------------|--|
| ● Doctors at multiple hospitals | <i>Dedicated DoctorAffiliation join table with an isPrimary flag</i> |
| ● Duplicate / spam voting | <i>Unique constraint on [reviewId, userId] in ReviewVote</i> |
| ● Optional sub-ratings | <i>Knowledge & bedside manner optional ; staff & punctuality mandatory</i> |
| ● Reviews spanning entity types | <i>Optional foreign keys —one review targets hospital, doctor, or service</i> |
| ● Privacy leakage in logs | <i>SystemLog.ipAddress stored in a masked, limited format</i> |
| ● Unmoderated content going live | <i>Every new review defaults to PENDING status</i> |

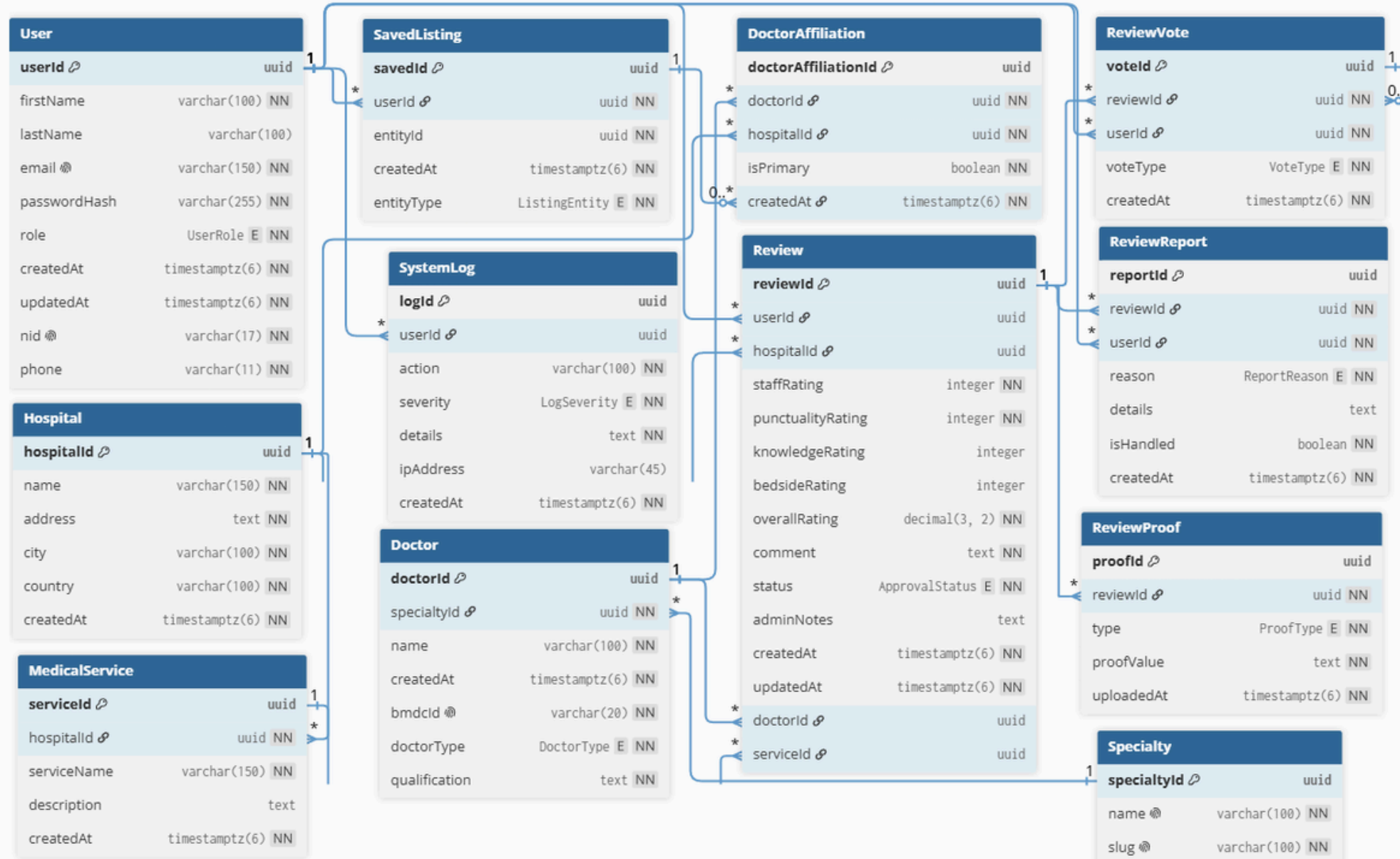
Request flow through the platform



EXAMPLE — REVIEW SUBMISSION



Core entities & relationships



SCHEMA

12 relational models, grouped by role

Core Entities	Relationship Entity	Engagement Entities	System Entity
User Hospital Doctor Specialty MedicalService	DoctorAffiliation	Review ReviewProof ReviewVote ReviewReport SavedListing	SystemLog

KEY ENUMS

- UserRole: USER, ADMIN
 - Proof Type: BILL, APPOINTMENT_NUMBER, FILE_UPLOAD
 - Report Reason: SPAM, OFFENSIVE, HIPAA_VIOLATION, FALSE_INFO, OTHER
- Approval Status: PENDING, APPROVED, REJECTED
 - Vote Type: HELPFUL, NOT_HELPFUL
 - Log Severity: INFO, WARNING, ERROR, SECURITY

Technology stack



Backend

Node.js, Epress.js 5



Database

PostgreSQL



ORM

Prisma + @prisma/adapters-pg



Auth & Security

JWT, bcrypt password hashing



Validation

Zod



Dev Tools

dotenv, nodemon, TypeScript types

- *Version control: Git & GitHub*
- *Hardware: standard development machine (no specialized hardware required)*

IMPLEMENTATION

How it was built

- 1 Designed the relational schema —12 interconnected tables
- 2 Connected PostgreSQL via Prisma Client + environment config
- 3 Built the Express server with JSON parsing & error handling
- 4 Implemented auth —bcrypt hashing, JWT login/logout
- 5 Built authMiddleware and adminMiddleware for access control
- 6 Developed CRUD for Hospitals, Doctors, Specialties, Services
- 7 Built the review system —proof upload, approval, voting
- 8 Added SavedListing bookmarking for users
- 9 Wrote a seed script for sample test data
- 10 Tested endpoints and verified role-based access control

Repository: github.com/Raihans-code/Med-review

What the platform's data can reveal

	Rating distribution by category	Comparing staff, punctuality, knowledge & bedside scores per hospital surfaces specific operational weaknesses.
	Review approval trends	Tracking APPROVED vs REJECTED ratios shows how much fake or low-quality content is filtered.
	Doctor affiliation patterns	Analyzing DoctorAffiliation records shows how many doctors practice across multiple hospitals.
	Report reason breakdown	Aggregating ReviewReport.reason values highlights the most common moderation issues.
	Helpfulness voting trends	ReviewVote data surfaces which reviews the community trusts most, independent of admin status.

RESULTS

What's working end to end



Authentication

Registration, login (JWT),
logout —all functional



Directory CRUD

Hospital / doctor /
specialty / service, admin-
protected



Proof-based reviews

Submission with
mandatory proof,
defaults to PENDING



Admin moderation

Approve / reject
endpoints with notes



Voting & reporting

Helpful/not-helpful votes and
report flow operational



Saved listings

Bookmarking functional for
authenticated users



Access control

Role-based routes verified
for admin vs. regular users

CONCLUSION

A trustworthy backbone for healthcare reviews

MedReview delivers a secure, relational backend for a proof-verified review platform — addressing the core trust problem through structured ratings, evidence-backed submissions, and admin moderation.

COMPLETED

- ✓ Full authentication & role-based authorization
- ✓ Complete relational schema (12 tables)
- ✓ CRUD for the full provider directory
- ✓ Proof-based reviews with moderation & voting
- ✓ Audit logging foundation via SystemLog

WHAT'S NEXT

Frontend client (web/mobile) for the API

Automated proof verification via OCR

Admin analytics dashboard with charts

Notification system for review status

Search & filter by rating and specialty